A dedicated roadmap for proving simple-even for A_λ

A self-contained LaTeX memorandum

Purpose

This document gives a focused roadmap for proving the *simple-even* property for the canonical selfadjoint operator

$$A_\lambda$$

attached to the restricted Weil quadratic form on the interval $[\lambda^{-1}, \lambda]$, for each fixed $\lambda > 1$. The goal is to organize the problem into proof routes, intermediate lemmas, fallback strategies, and addenda worth preserving for robustness and future generalization.

The central point is that this problem should not be approached as a single monolithic theorem. It naturally breaks into four layers:

1. canonical realization and compact resolvent for the full form;
2. structural control of a simpler base operator L_λ;
3. transfer of ground-state information from L_λ to

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda,$$

   where C_λ is scalar and K_λ is finite-rank or very low complexity;
4. parity and rigidity arguments that turn uniqueness of the ground state into evenness.

The document is written to support three uses at once:

- as a proof-planning memo for the current paper;
- as a checklist for targeted literature mining;
- as a design document for later generalization to Dedekind zeta functions or automorphic L-functions.

1. Target statement and stronger variants

Primary target

For each fixed $\lambda > 1$, let A_λ be the canonical lower bounded selfadjoint operator associated with the restricted Weil quadratic form QW_λ. The primary target is:

> **Simple-even target.** The lowest eigenvalue $\varepsilon(\lambda) = \min \operatorname{spec}(A_\lambda)$ is simple, and an associated eigenfunction can be chosen even under reflection.

Since A_λ has compact resolvent in the CCM framework, the bottom of the spectrum is always an eigenvalue. Thus the problem is not existence of a lowest eigenvector, but its *multiplicity* and *parity*. Connes' 2026 survey explicitly isolates this as one of the remaining steps in the restricted-Weil-form strategy toward RH [4]. The 2025 paper *Zeta Spectral Triples* also treats the corresponding finite-dimensional truncations under an explicit even-simple assumption [3].

Useful stronger variants

Several stronger statements would be even better than simple-even and are worth keeping visible.

(S1) **Strict positivity.** The lowest eigenfunction is strictly positive a.e. on the logarithmic interval.

(S2) **Simplicity on the even sector.** The bottom of A_λ restricted to the even subspace is simple, and lies strictly below the bottom of the odd sector.

(S3) **Analytic continuation in a perturbation parameter.** For

$$A_{\lambda,t} := L_\lambda + C_\lambda I + tK_\lambda, \qquad t \in [0,1],$$

the lowest eigenvalue remains simple for all t, with analytic eigenbranch.

(S4) **Uniform gap statement.** There exists a positive gap between the lowest and second eigenvalues on the even sector, either for each fixed λ or on a useful λ-range.

A proof of any stronger form can be used to deduce the primary target. In practice, it is often easier to prove strict positivity or a sector-gap inequality than to attack simplicity head-on.

2. Known input and what it buys

The roadmap assumes the following background inputs are available or at least strongly supported by the current paper and the CCM literature.

(K1) The restricted Weil quadratic form is densely defined, lower bounded, and closed, hence determines a canonical selfadjoint operator A_λ by the representation theorem for semibounded forms [3, 10].

(K2) The operator A_λ has compact resolvent, so its spectrum is discrete and the lowest spectral value is an eigenvalue [3].

(K3) The current paper provides an explicit decomposition of the full form into a base part plus lower-order corrections, leading formally to

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda.$$

The scalar shift $C_\lambda I$ does not affect multiplicities or eigenvectors, so all difficulty lies in transferring ground-state information from L_λ through K_λ.

(K4) The operator commutes with reflection. Hence each eigenspace decomposes into even and odd parts.

(K5) The CCM program gives two additional pieces of input: first, a canonical formulation of A_λ for the full form; second, a truncated spectral-triple package whose selfadjointness and real-zero conclusions are available once an even-simple hypothesis is supplied [2, 3, 5].

These known facts reduce the simple-even problem to a much sharper question: *how can one prove uniqueness and parity of the ground state in the presence of a finite-rank correction to a more positivity-friendly base operator?*

3. Obstacle map

The main obstructions should be stated plainly.

Obstacle A: the full form need not be Markovian

Even if the principal/base part has Dirichlet-form or positivity-preserving structure, the full operator A_λ can fail to generate a positivity-preserving semigroup because of the low-rank arithmetic correction. Thus Perron-Frobenius is unlikely to apply directly to the full operator without extra work.

Obstacle B: finite rank does not automatically preserve the ground-state package

A bounded or finite-rank perturbation preserves selfadjointness and discreteness, but it does *not* automatically preserve positivity of the ground state or simplicity of the lowest eigenvalue. Therefore one needs either smallness, monotonicity, sector reduction, or a determinant/Birman-Schwinger argument at the bottom of the spectrum.

Obstacle C: evenness is not independent of simplicity

If the lowest eigenspace has dimension greater than one, reflection symmetry alone does not force evenness. The standard implication is:

$$\text{uniqueness of a positive ground state} + \text{reflection symmetry} \Longrightarrow \text{ground state is even.}$$

So in practice one should try to prove *simplicity first*, then deduce evenness almost for free.

Obstacle D: different λ-regimes may require different tools

Small λ, intermediate λ, and asymptotically large λ may not yield to the same proof mechanism. For example, large-λ behavior may be approachable through prolate or semiclassical asymptotics, while fixed moderate λ may be better handled by abstract perturbation theory or numerically guided variational bounds. The roadmap should therefore allow more than one proof route.

4. Primary route: prove a full ground-state package for L_λ and transfer it to A_λ

This is the preferred route because it aligns with the structural gains already obtained in the current paper.

4.1. Stage I: prove the base operator has the right positivity structure

The first target should be to prove, for the closed form associated to L_λ, some package close to the following:

(I.1) symmetry and lower boundedness;

(I.2) closedness on a natural domain;

(I.3) positivity preservation of the semigroup e^{-tL_λ};

(I.4) ideally the Markov property, or at least enough Beurling-Deny structure to invoke standard positivity theorems [6, 9].

What to mine. The relevant literature is the symmetric Dirichlet-form and positive-semigroup spine: Fukushima–Oshima–Takeda for closed symmetric Markov forms and invariance criteria [6]; Ouhabaz for positivity preservation, irreducibility, and domination arguments [9]; and Keller–Lenz–Vogt–Wojciechowski for positivity-improving semigroups and their spectral consequences [8].

Why it matters. If L_λ generates a positivity-improving semigroup and has compact resolvent, then the lowest eigenvalue of L_λ is simple and the ground state is strictly positive. This is the cleanest possible starting point for transferring simplicity to the full operator.

4.2. Stage II: prove irreducibility or positivity improvement for L_λ

Positivity preservation alone is often not enough; one wants irreducibility or positivity improvement. There are several possible strategies.

(II.1) **Kernel-connectivity route.** Show that the jump kernel or difference-energy kernel connects every nontrivial measurable subset of the interval to its complement, ruling out invariant ideals.

(II.2) **Invariance-criterion route.** Use the classical theorem that irreducibility of a symmetric Dirichlet form is equivalent to absence of nontrivial invariant measurable subsets under the semigroup [6, 9].

(II.3) **Integral-operator route.** If the resolvent or semigroup has a strictly positive kernel for positive time, positivity improvement follows directly.

Deliverable. A theorem of the form: "For each fixed $\lambda > 1$, the semigroup generated by L_λ is positivity improving on $L^2(I_\lambda)$." This would immediately imply simplicity and strict positivity of the L_λ ground state.

4.3. Stage III: isolate the even sector already at the base level

Because reflection is built in, it is useful to decompose

$$L^2(I_\lambda) = L^2_{\text{even}} \oplus L^2_{\text{odd}}.$$

Then one can ask whether the base operator's ground state is already known to lie in the even sector. There are two natural ways to prove this.

(III.1) **Positivity route.** A strictly positive eigenfunction cannot be odd, so positivity improvement automatically forces evenness once reflection commutation is known.

(III.2) **Energy comparison route.** Show directly that odd functions pay more energy than their even counterparts, for example by comparing the form on symmetric pairs.

Recommendation. Keep the positivity route as the main line. The energy-comparison route is valuable as an addendum because it may extend to contexts where positivity improvement is weaker or unavailable.

4.4. Stage IV: transfer simplicity from L_λ to A_λ

This is the critical step. It likely needs more than one proof technique.

Route IV-A: analytic continuation in a perturbation parameter

Consider

$$A_{\lambda,t} = L_\lambda + C_\lambda I + tK_\lambda, \qquad 0 \le t \le 1.$$

If K_λ is bounded selfadjoint and the bottom eigenvalue of L_λ is simple and isolated, then Kato theory gives an analytic eigenbranch for small t [7]. To extend simplicity all the way to $t = 1$, one needs a mechanism preventing eigenvalue crossing.

What must be shown.

(A1) the lowest eigenvalue of $A_{\lambda,t}$ remains isolated for all $t \in [0, 1]$;

(A2) no odd eigenbranch crosses the even ground-state branch;

(A3) no second even eigenbranch collides with the ground-state branch.

How to do it. This suggests proving a lower bound on the gap either on the whole space or separately on the odd sector. Min-max and parity decomposition are likely the natural tools.

Route IV-B: Birman-Schwinger or low-rank determinant method

Because K_λ is finite-rank or very low rank, one can try to reduce the perturbed eigenvalue problem

$$(A_\lambda - \mu)f = 0$$

to a finite-dimensional determinant equation involving the resolvent of the base operator:

$$\det\big(I + K_\lambda(L_\lambda + C_\lambda I - \mu)^{-1}\big) = 0.$$

This is a strong route because it turns multiplicity questions near the bottom of the spectrum into low-dimensional algebra. Abstract versions of this viewpoint are standard in Birman-Schwinger theory and low-rank perturbation theory [1].

What to mine. Extract an explicit matrix model for the perturbation on the span of the low-rank vectors. Then study monotonicity of the determinant near the base ground-state energy. If one can show there is exactly one zero below the rest of the spectrum, simplicity follows.

Route IV-C: sector-by-sector min-max

Because of reflection symmetry, prove separately that

$$\inf \operatorname{spec}(A_\lambda|_{\text{even}}) < \inf \operatorname{spec}(A_\lambda|_{\text{odd}}).$$

Then prove the even-sector bottom eigenvalue is simple. This route is especially attractive when the odd sector is easier to control because odd functions are forced to vanish at the center in logarithmic coordinates, or because the perturbation couples more naturally to even modes.

Recommendation. This route should be developed in parallel with IV-A. It may be the most realistic mechanism for preventing parity-changing crossings in the Kato continuation picture.

4.5. Stage V: deduce evenness

Once simplicity of the ground state is known, evenness should be proved as a short corollary:

> If A_λ commutes with reflection and its lowest eigenspace is one-dimensional, then the corresponding eigenfunction is an eigenvector of reflection. If one also knows positivity, it must be the even one.

Even without positivity, one can still prove the eigenvector is either even or odd. Then one only needs to rule out oddness, preferably through an energy or nodal argument.

5. Secondary route: attack the full operator directly

The base-operator-first route is preferred, but a direct route should also be developed because it may reveal shortcuts.

5.1. Direct form route

Attempt to prove that the *full* restricted Weil form is positivity preserving after a suitable ground-state transform, or at least dominated by a positivity-improving comparison form. This is harder,

because low-rank terms often break the Markov property, but it can still be worthwhile if the perturbation has a definite sign or factors through positive functionals.

5.2. Direct parity route

One can try to show directly that the minimizer of the full form must be even. Natural tools include:

1. symmetrization with respect to reflection;
2. comparing $QW_\lambda(f, f)$ with $QW_\lambda(\frac{1}{2}(f + Rf), \frac{1}{2}(f + Rf))$;
3. showing the odd part contributes nonnegatively or strictly positively.

This route may fail if the perturbation mixes parities in a subtle way, but if it succeeds it would bypass the hardest transfer step.

5.3. Direct nodal route

On compact-resolvent selfadjoint operators, the ground state often has a fixed sign. If a version of the Courant nodal principle or sign-definiteness can be established directly for the full operator, then simplicity becomes much more accessible. This is less standard for nonlocal operators, but not impossible.

Recommendation. Keep this as a secondary line of attack. It is less structurally transparent than the base-operator route, but it may produce concise arguments in the even sector.

6. CCM-assisted route: use truncated and prolate structures as guides

The CCM papers do not solve simple-even for A_λ, but they provide two useful ingredients.

6.1. Truncated even-simple package

Zeta Spectral Triples works with finite-dimensional truncations of the Weil form and explicitly assumes an even-simple property for the smallest eigenvalue of the truncated matrix to obtain selfadjointness of the perturbed scaling operator and the real-zero property of the associated Fourier transform [3]. This suggests a finite-dimensional diagnostic program:

(T1) study even-simple numerically for truncated matrices across many λ and truncation levels N;

(T2) extract monotonicity or interlacing patterns in the truncated spectra;

(T3) try to pass these patterns to the operator limit.

This does not replace a proof, but it may reveal the right finite-rank determinant identities or the right gap estimates to prove in the continuum.

6.2. Prolate/quasimode guidance

Connes' 2026 survey identifies prolate-wave constructions as educated approximations to the eigenvector corresponding to the smallest eigenvalue of A_λ and states that the analogue of simple-even is known for the prolate operator [4]. This suggests the following hybrid strategy:

(P1) build a family of explicit quasimodes k_λ in the even sector;

(P2) prove they have Rayleigh quotient within a controlled distance of the true ground-state energy;

(P3) use these quasimodes to get a quantitative upper bound for the even-sector bottom;

(P4) combine with lower bounds on the odd-sector bottom to force the true ground state into the even sector.

This route is especially promising if a direct positivity-improving theorem for A_λ is unavailable.

6.3. Determinant viewpoint

CCM's determinant formulas may also suggest a way to control multiplicities. If the ground-state eigenfunction or its truncations yield entire functions with real zeros that encode the spectrum of a companion operator, then multiplicity information may be reflected in zero multiplicities or rank defects. This is speculative, but worth monitoring as a side route.

7. Number-theoretic inputs that can strengthen the roadmap

Although simple-even is primarily an operator-theoretic problem, several number-theoretic inputs may sharpen the proof.

Explicit formulas and sign structure

A more refined decomposition of the prime and gamma-factor terms can reveal whether some components of the perturbation have a favorable sign, are rank-one or rank-two, or vanish on odd functions. Such information can make a parity comparison argument much sharper.

Prime number theorem type bounds

Effective control of prime sums and prime-power tails can yield explicit estimates on the size of the nonprincipal correction terms. This is relevant if one wants to prove that the perturbation is small compared with the base spectral gap in certain regimes of λ.

Gamma identities and duplication formulas

Archimedean terms sometimes simplify dramatically after duplication/reflection identities for the gamma function. If these identities reduce the apparent complexity of scalar or low-rank corrections, they can make perturbation arguments more transparent and improve estimates.

Trace and product formulas

Trace identities or product formulas may provide independent information on low eigenvalues, especially if one can compare the trace of resolvents or heat semigroups for L_λ and A_λ. This is not the first route to try, but it could be effective once a candidate gap inequality is in hand.

8. What probably needs multiple proof techniques

A single proof method is unlikely to settle the entire problem. The following splits are realistic.

Simplicity versus evenness

Simplicity is likely best attacked by positivity improvement, irreducibility, spectral gap inequalities, or low-rank determinant arguments. Evenness is likely best deduced afterward from reflection symmetry and either positivity or sector comparison.

Fixed λ versus all λ

For each fixed λ, an operator-theoretic proof may be enough. For a statement uniform in λ or valid across all regimes, one may need to combine:

- abstract semigroup methods for structural facts;
- perturbative continuation in t for transfer from L_λ to A_λ;
- asymptotic or prolate methods for large λ;
- finite-dimensional or numerical guidance for moderate λ.

Base operator versus full operator

The base operator is the natural home for Dirichlet-form and positivity methods. The full operator is the natural home for canonicality and the exact Connes/CCM statement. A complete solution may therefore require one argument to prove a strong theorem for L_λ and a second argument to transfer it through K_λ.

9. Concrete milestone plan

The following sequence is designed to produce usable progress even if the full target is not reached immediately.

Milestone	Statement to prove	Main tools	Why it matters

M1	Closedness and compact resolvent for L_λ in a clean abstract package	Form methods, compact embedding, interval compactness	Makes the base operator usable as the main comparison object
M2	Positivity preservation and irreducibility for L_λ	Beurling-Deny, invariance criteria, kernel connectivity	Gives simple positive ground state for L_λ
M3	Ground state of L_λ is even and simple	Positivity + reflection symmetry	Establishes the target package at base level
M4	Low-rank determinant formula for A_λ relative to L_λ	Birman-Schwinger, finite-rank perturbation theory	Converts the transfer problem into finite-dimensional algebra
M5	The even-sector ground state of A_λ lies below the odd sector	Min-max, parity decomposition, quasimodes	Prevents parity-changing crossings
M6	Simplicity of the A_λ ground state	Kato continuation plus gap control, or determinant monotonicity	Gives the full simple-even theorem
M7	Optional strengthening: analytic dependence and gap persistence in t or λ	Analytic perturbation theory	Upgrades the result and supports generalization

10. Recommended addenda for robustness and future flexibility

This roadmap should be supported by addenda, not because the main proof should be bloated, but because several alternate arguments are strategically valuable.

Addendum A: alternate irreducibility proof for L_λ

Include both a semigroup/invariance proof and a kernel-connectivity proof if available. The first is elegant; the second is often easier to adapt when the explicit formula changes.

Addendum B: alternate compactness proof

Preserve both a direct interval compactness proof and an abstract form-embedding proof if they are both available. Different future L-functions may favor different versions.

Addendum C: parity comparison lemmas

Collect any lemmas showing that odd functions pay extra energy, or that the perturbation is simpler on even functions. These may become central if positivity-improving transfer to the full operator turns out to be hard.

Addendum D: finite-rank algebra

Write out the resolvent identities, matrix reductions, and determinant formulas for the low-rank perturbation in full detail. This is important both for rigor and for numerics.

Addendum E: prolate/quasimode package

Include the precise quasimode constructions, their Rayleigh-quotient bounds, and any overlap estimates with the true ground state. Even if not needed in the final proof, this material is likely to remain useful.

11. Recommended order of work

The best order is not to attack the full theorem directly. The recommended sequence is:

1. Finish the clean abstract package for L_λ: closed form, selfadjoint realization, compact resolvent.
2. Prove positivity preservation and irreducibility for L_λ.
3. Deduce simple positive even ground state for L_λ.
4. Build the low-rank transfer machinery for A_λ.
5. In parallel, develop sector-by-sector bounds and CCM-guided quasimodes.
6. Only then attempt the final transfer to the full operator.

This order maximizes the chance of obtaining publishable intermediate results even if the full simple-even statement for A_λ remains open for a time.

12. Bottom line

The most realistic path to proving simple-even for A_λ is not a direct attack on the full restricted Weil form. It is a *hybrid strategy*:

> prove a strong ground-state theorem for the base operator L_λ by positivity and compactness methods, then use low-rank perturbation theory, parity decomposition, and CCM-inspired quasimodes to transfer the conclusion to the canonical full operator A_λ.

The decisive technical hinge is therefore the transfer step. Everything in the current paper and the surrounding literature should be mined with that hinge in mind.

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